

Start your Fitness Journey

Start your fitness journey
with our app's guidance and support.



Lets start



DEVELOPMENT OF WEB AND MOBILE APPLICATIONS FOR AI- BASED PERSONALIZED WORKOUT AND NUTRITION PLANNING

Team memebers: Arlen Baibossyn, Nurbol
Nurlybay, Muslim Toleumukahanbet

Supervisor: Olzhas Shaiken

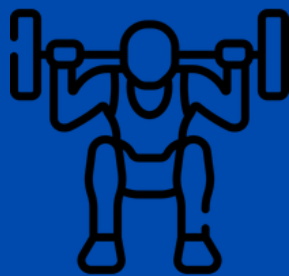
PROBLEM



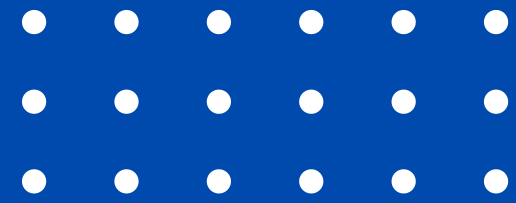
Students and professionals struggle to stay healthy due to lack of time, money, and consistency.



Nutrition planning is difficult, so people don't know what or how much to eat especially beginner and intermediate fitness users.



Many apps are too generic and don't explain recommendations, so users quit. Consequently, these users prefer guided digital fitness solutions that provide personalized coaching .



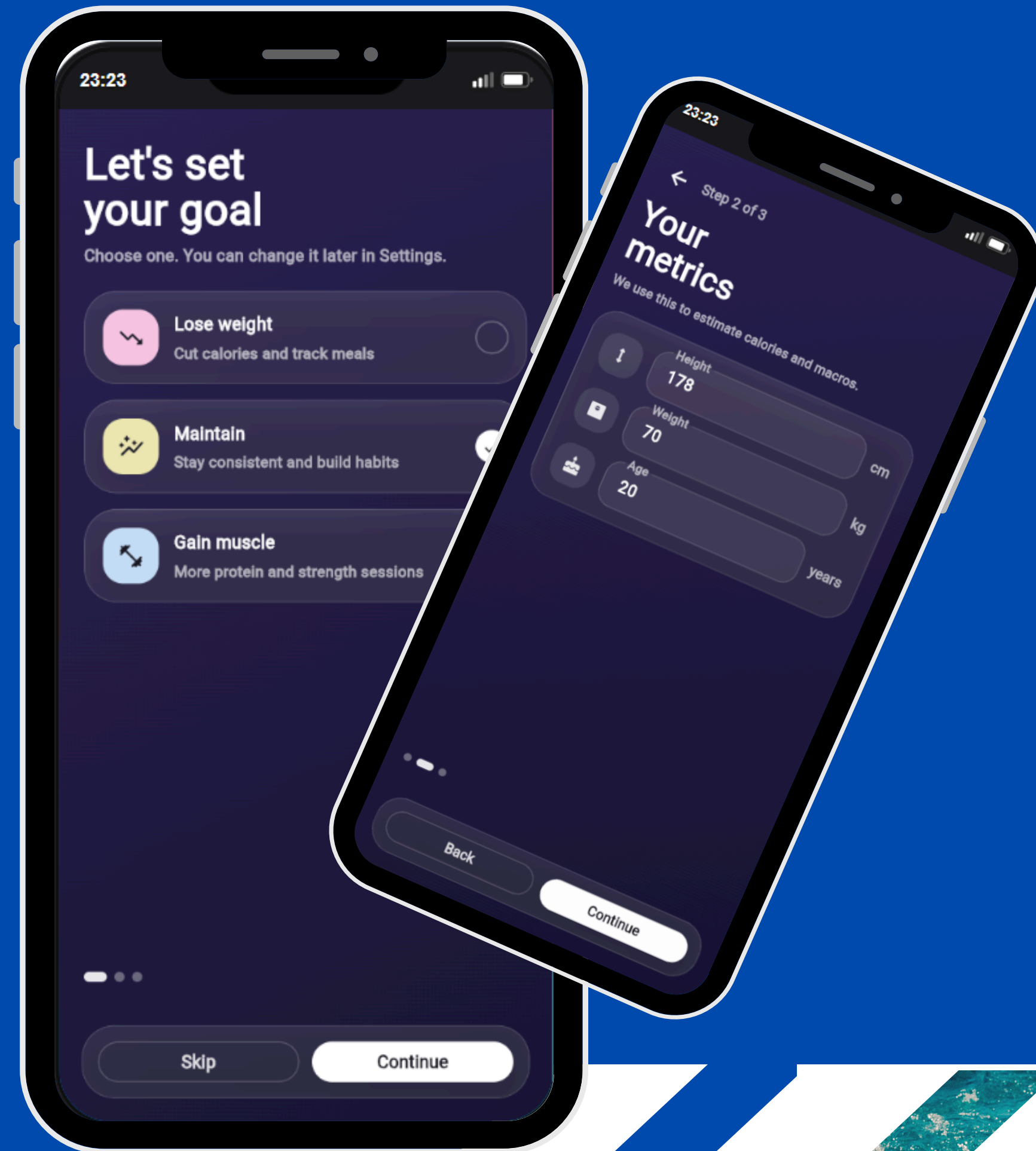
TARGET AUDIENCE

The MVP primarily targets:

- University students (18–30 years)
- Young professionals with busy schedules

The audience was narrowed down to individuals who:

- Own a smartphone
- Prefer guided digital fitness solutions
- Seek affordable, flexible, and personalized fitness support



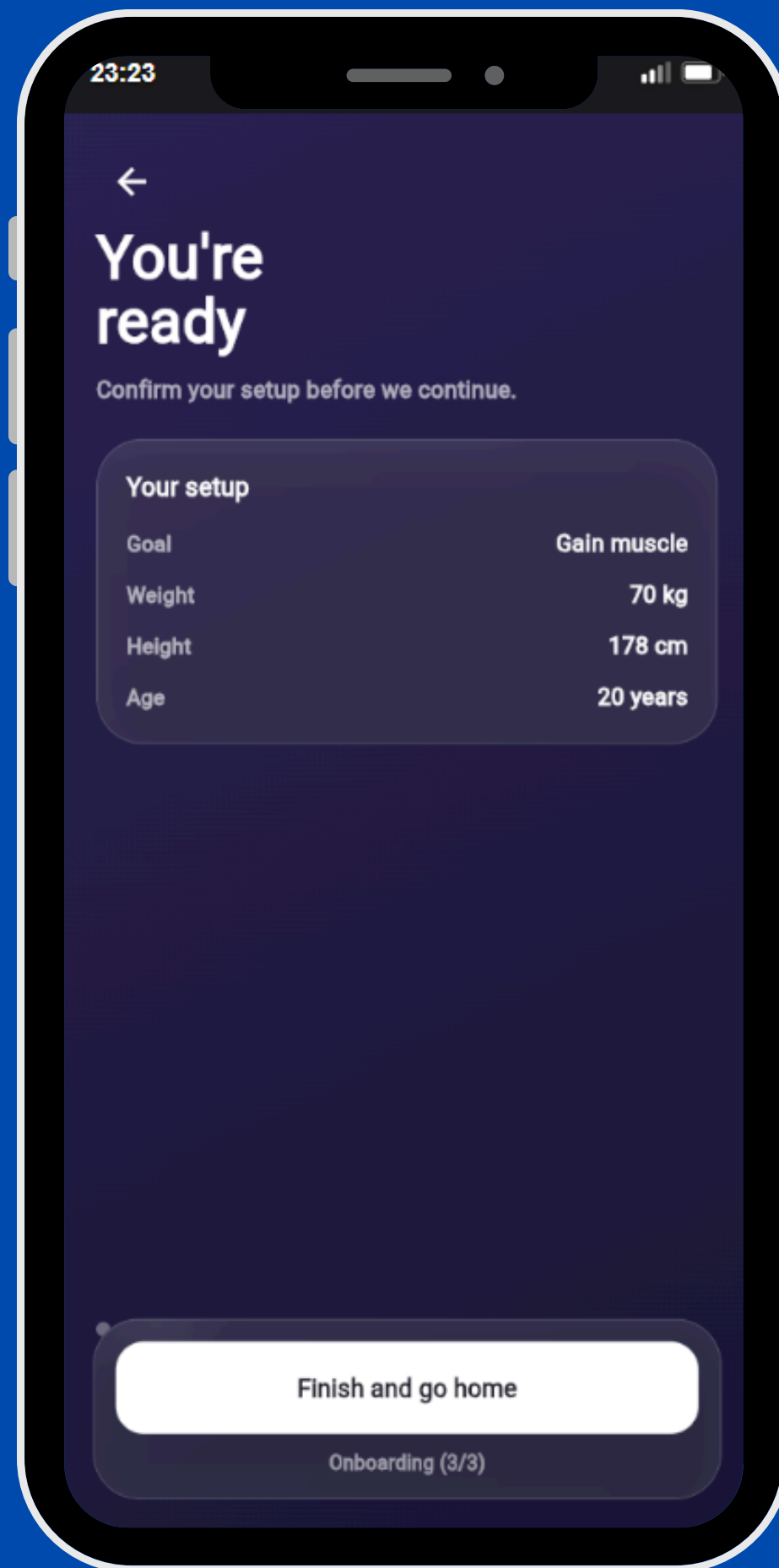
COMPETITORS

Why our app is unique

Short comparison (2 competitors) — meal tracking + workouts + AI photo analysis

Feature	Our App	MyFitnessPal	Nike Training Club
Meal tracking	Photo + manual entry	Manual food log	Not included
Workouts	Plans + video demos	Limited workouts	Workouts + videos
AI analysis	Calories + macros estimate	No photo AI	No nutrition AI
All-in-one flow	Meal → Analyze → Save	Food tracking only	Workouts focus

Note: competitor feature sets vary by region and version.



SOLUTION OVERVIEW

- AI-Based Fitness Planning
- Adaptive Training Sessions
- AI-Generated Exercise Videos
- Nutrition Tracking
- Progress and Profile Management

METHODOLOGY / TECHNOLOGIES

METHODOLOGY

- step-by-step development: analysis, design, implementation, testing.

FLUTTER / DART

- mobile application development.

FIREBASE

- authentication and data storage.

AI

- personalized workout plans and exercise videos.

RIVERPOD / GOROUTER

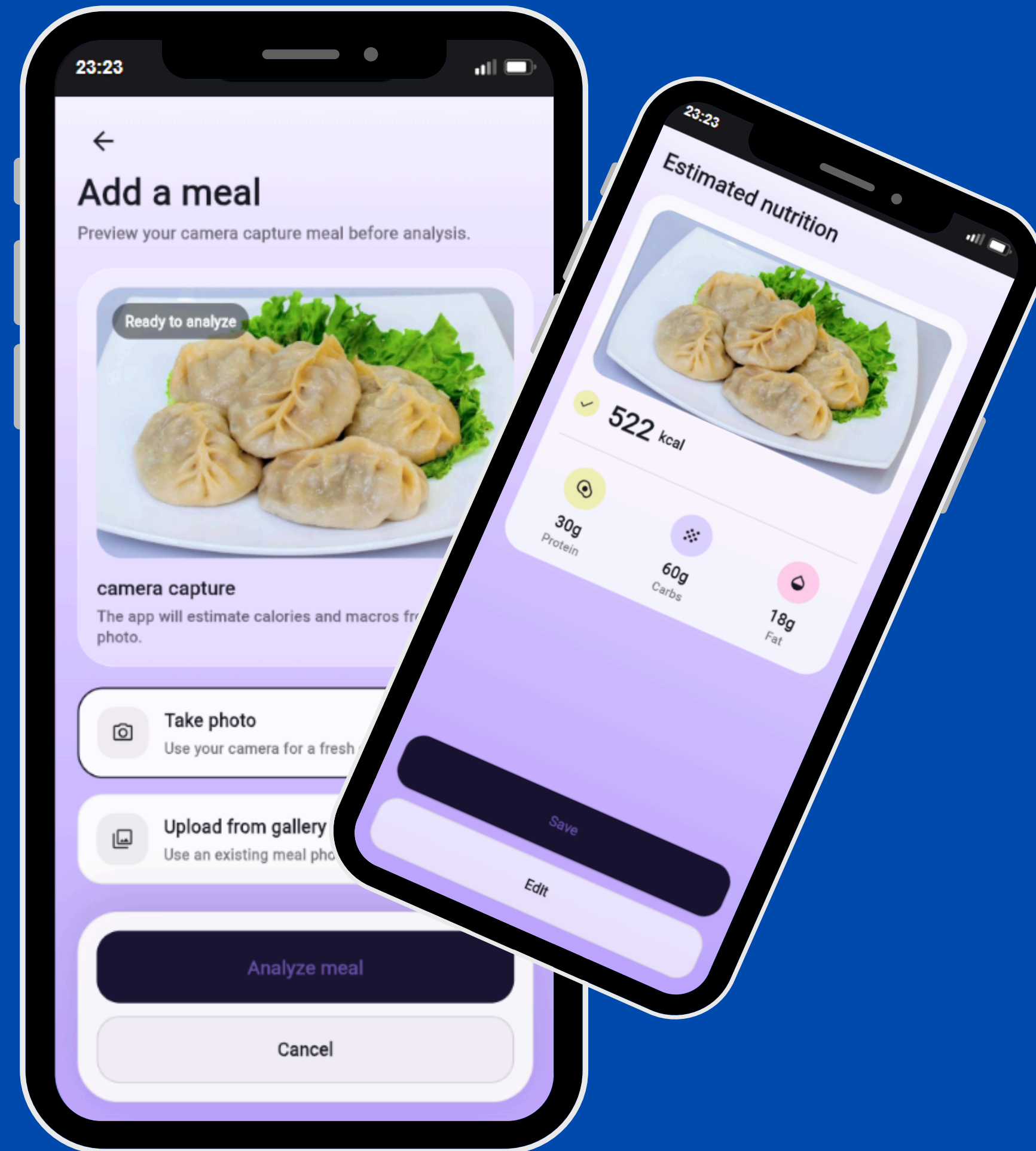
- state management and navigation.

MEAE TRACKING

MEAL PHOTO → NUTRITION ESTIMATION



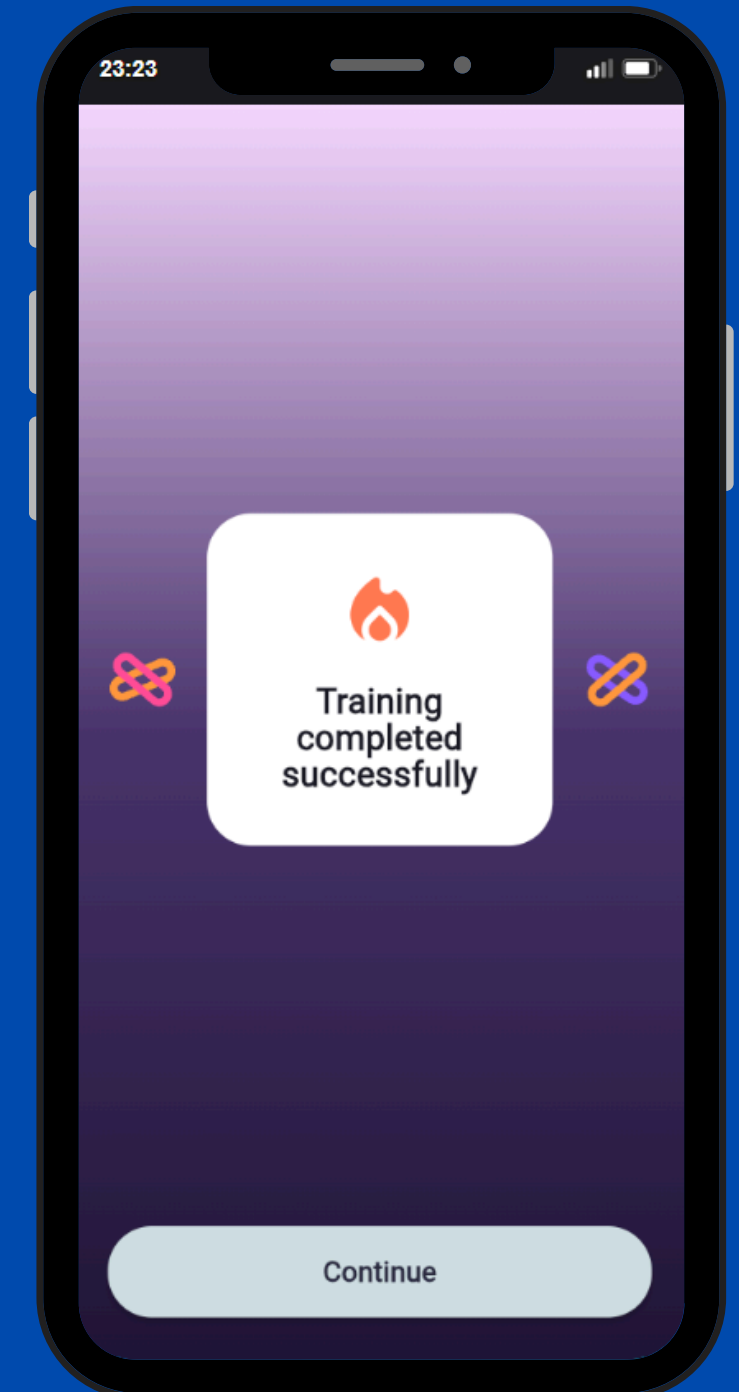
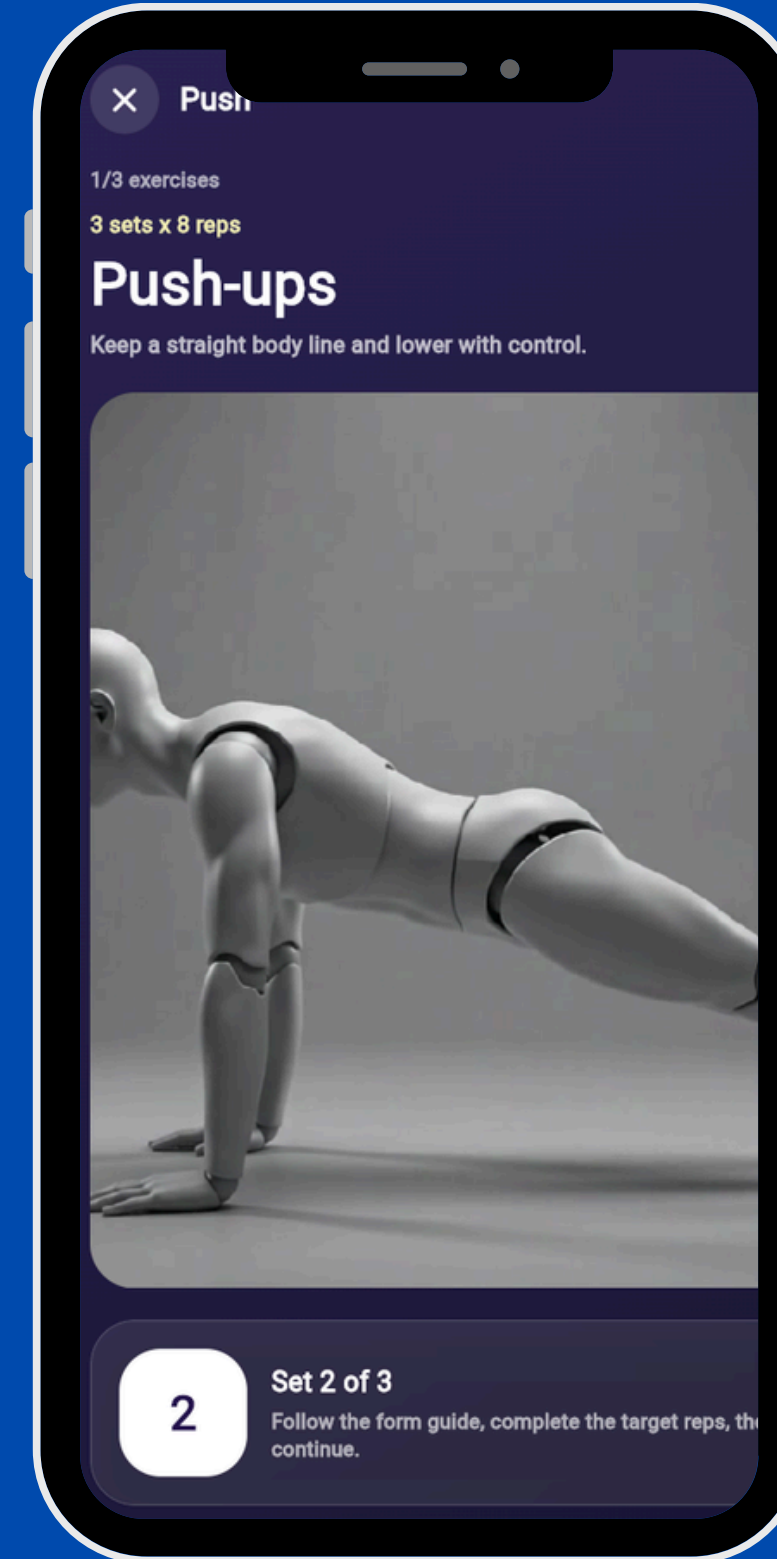
- Take/upload meal photo
- Nutrition API estimates calories and macros
- User can edit and save



AI-GUIDED TRAINING SESSION

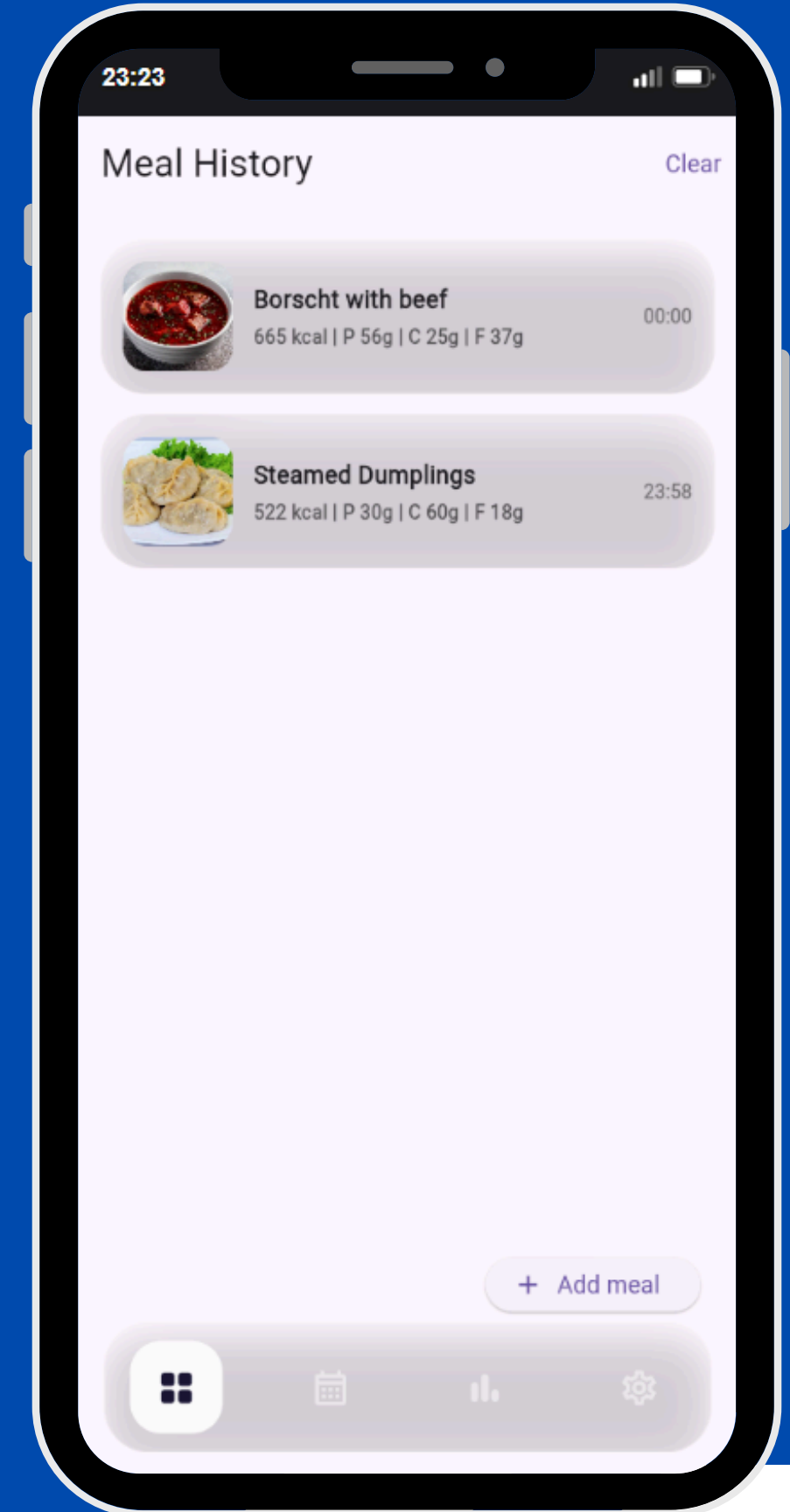
PERSONALIZED WORKOUT GENERATED BY AI

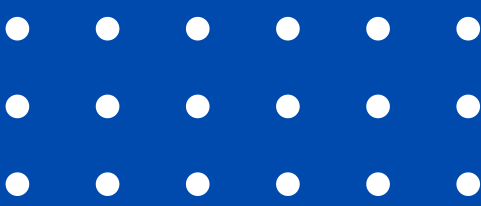
- **TIMER-GUIDED EXERCISES WITH WORK/REST INTERVALS**
- **REAL-TIME SESSION TRACKING**
- **AUTOMATIC COMPLETION AND PROGRESS UPDATE**



MAIN IMPLEMENTATION PRIORITY

- **Collects user profile data during onboarding.**
- **Generates a personalized workout plan by goal.**
- **Uses AI-generated videos for exercise guidance.**
- **Supports reps, sets, timers, and rest periods.**
- **Stores progress and nutrition data in Firebase.**





CONCLUSION

The project resulted in a functional AI-based fitness app with personalized workouts, exercise video guidance, nutrition tracking, and Firebase data storage. Overall, the system demonstrates that AI can make fitness training more adaptive, visual, and user-focused.